「Normal and Hermitian.

「Lemma. Let V be an inn. Prod space and T is a Hermitian operator.
(If exists) Every eigenvalue of T is real.
Moreover, eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue c of T corresponding to α,

$$c\cdot\langle\alpha,\alpha\rangle = \langle T(\alpha),\alpha\rangle = \langle\alpha, T^*(\alpha)\rangle = \langle\alpha, T(\alpha)\rangle = \langle\alpha, c\cdot\alpha\rangle = \bar{c}\cdot\langle\alpha,\alpha\rangle$$

Since α is non-zero, $c=\bar{c}$, or c is a real number.
And, take α_1,α_2 be eigenvectors corresponding to c_1, c_2, with $c_1\neq c_2$.
Then,

$$c_1\cdot\langle\alpha_1,\alpha_2\rangle = \langle c_1\alpha_1,\alpha_2\rangle = \langle T(\alpha_1),\alpha_2\rangle = \langle\alpha_1, T^*(\alpha_2)\rangle = \langle\alpha_1, T(\alpha_2)\rangle = \langle\alpha_1, c_2\alpha_2\rangle = \bar{c}_2\cdot\langle\alpha_1,\alpha_2\rangle$$

Since c_2 must be real, $\bar{c}_2 = c_2$. Now, since $c_1\neq c_2$,

$$(c_1-c_2)\cdot\langle\alpha_1,\alpha_2\rangle = 0 \text{ implies } \langle\alpha_1,\alpha_2\rangle = 0.$$

「Thm. In finite-dimensional inn. prod space,
Every Hermitian operator has an eigenvector.

Proof. If the given field is $\mathbb{C}$, then the charateristic polynomial must have a root in $\mathbb{C}$
by the Fundamental Theorem of Algebra. So it has an eigenvector.
Now, assume that the given field is the $\mathbb{R}$.
Let T be a Hermitian on V over $\mathbb{R}$, and β be an orthonormal basis for V.
Let $A=[T]_\beta$. Since $T=T^*=T^t$, $A=A^*=A^t$. Define an inner product on $\mathbb{C}^n$ as:

$$\langle X,Y\rangle \overset{\text{def}}{=} Y^*X$$

Then, a linear operator $L_A:\mathbb{C}^n\to\mathbb{C}^n : X\mapsto AX$ is Hermitian, because

$$\langle L_A(X),Y\rangle = \langle X, L_A^*(Y)\rangle \text{ and } \langle AX,Y\rangle = Y^*AX = \langle X, A^*Y\rangle = \langle X, AY\rangle.$$

By the FTA, L_A has an eigenvalue, and that must be real, by the above Lemma.
Now, $L_A(X)=AX=cX$ for some real number c and a nonzero vector X in $\mathbb{C}^n$,
or $\det(cI-A)=0$, $A-cI$ is not invertible, and $A-cI$ has only real-entries,
so $(A-cI)X=0$ has a non-zero real solution X.

「Lemma. Let V be a fin. dim inn. Prod space, and T be a linear operator on V.
If $W<V$ is T-invariant, then $W^\perp$ is T^*-invariant.

Proof. Given $\beta\in W^\perp$, and $\alpha\in W$, $\langle T^*(\beta),\alpha\rangle = \langle\beta, T(\alpha)\rangle = 0$, by $T(\alpha)\in W$. ▨

「Diagonalizable Criterion of linear operator on inner product space.

「Thm. Let T be a linear operator on fin. dim inn. prod space V.
If T is Hermitian, then there is an orthonormal basis for V consisting by eigenvectors of T.

proof. Let T be a Hermitian. Using induction for $\dim V = n$.
If $n=1$, trivial. And, for n, using the facts that:
1). T has an eigenvector α_1.
2). $\langle \alpha_1 \rangle \oplus \langle \alpha_1 \rangle^{\perp} = V$.
3). $\langle \alpha_1 \rangle$ is T-invariant, so $\langle \alpha_1 \rangle^{\perp}$ is T^*-invariant, and $T = T^*$.

「Corollary. In $\mathbb{R}$-Inner Product space,
T is symmetric <u>if and only if</u> there is orthonormal basis for V consisting by eigenvectors of T.

「Lemma. Let T be a normal operator on fin. dim inn. prod space V. Given $\alpha \in V$,
α is eigenvector for T corresponding to $c \in F$ <u>iff</u> α is eigenvector for T^* corresponding to $\bar{c} \in F$.

proof. If L is normal, then
$$\|L(\alpha)\|^2 = \langle L(\alpha), L(\alpha) \rangle = \langle \alpha, L^*L(\alpha) \rangle = \langle \alpha, LL^*(\alpha) \rangle = \langle L^*(\alpha), L^*(\alpha) \rangle = \|L^*(\alpha)\|^2.$$
And, any scalar $c \in F$, $T - cI$ is normal, by
$$(T-cI)(T-cI)^* = TT^* - \bar{c}T - cT^* + c\bar{c} = T^*T - \bar{c}T - cT^* + c\bar{c} = (T-cI)^*(T-cI).$$
So, $\|(T-cI)(\alpha)\| = \|(T-cI)^*(\alpha)\|$ gives the result.

「Thm. Let T be a linear operator on fin. dim V.S V, and $\mathcal{B}$ be an orthonormal basis for V.
Suppose that $A = [T]_{\mathcal{B}}$ is upper-triangular. Then,
T is normal <u>if and only if</u> A is diagonal.

proof. Let T is normal and $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$. Since A is upper triangle,
$$T(\alpha_1) = A_{11}\alpha_1 \text{ and so } T^*(\alpha_1) = \bar{A}_{11}\alpha_1 = \sum_{j=1}^{n} (A^*)_{j1} \cdot \alpha_j = \sum \bar{A}_{1j} \cdot \alpha_j$$
So, $\bar{A}_{12} = \bar{A}_{13} = \cdots = \bar{A}_{1n} = 0$. Similarly, all non-diagonal entries are zero.
The Converse is trivial. ☒
Now, since every linear operator on $\mathbb{C}$-vector space has a basis s.t the representation is triangular
Using Gram-Schmidt process to this, we obtain that every normal in $\mathbb{C}$ is unitary diagonalizable.

「Summary.

$T : V \to V$, V fin. dim inn. Prod over F

T is normal and $F = \mathbb{C}$

$\Updownarrow$

In $\mathbb{C}$, T is Hermitian $\Rightarrow$ T is unitary equivalent to a diagonal matrix

$\not\Downarrow$ $\not\Uparrow$

In $\mathbb{R}$, T is symmetric $\Leftrightarrow$ T is orthogonal equivalent to a diagonal matrix

$\not\Leftarrow$

In $\mathbb{C}$, Hermitian $\Rightarrow$ Normal

$\not\Downarrow$ $\not\Uparrow$

Symmetric